

Padanam AI — A Personalized Learning Agent for Kerala School Students

Padanam AI is an intelligent tutoring agent that adapts curriculum content to each student's learning speed, language preference (English/Malayalam), and topic-level mastery. It continuously senses student performance, reasons over an internal learner model, and acts by adjusting lesson difficulty, language, and feedback, closing the loop that static video/PDF tools and generic chatbots both miss.

2. PEAS Table

Component	Description
Performance measure	Growth in per-topic mastery scores, quiz accuracy improvement over time, appropriate difficulty/language match to the student, and positive engagement
Environment	The student, the curriculum content, quiz item bank, the student's grade/board, and the digital platform
Actuators	Displaying explanations, generating and presenting quizzes, adjusting difficulty level, switching explanation language, issuing study-plan recommendations, sending motivational feedback and notifications
Sensors	Quiz answers, response time and retries, chat messages (including confusion signals), language selection, navigation/click behaviour, historical progress records

3. Environment Properties

Property	Classification	Justification
Fully Observable vs Partially Observable	Partially observable	The agent cannot directly perceive a student's true understanding; it only infers mastery from proxies like quiz answers, hesitation time, and retries
Multi-Agents vs Single Agent	Single-agent	From the tutoring agent's own decision-making perspective, it acts alone; teachers/parents are environment stakeholders it reports to, not competing or cooperating agents it negotiates with
Deterministic vs Stochastic	Stochastic	The same explanation given twice to the same mastery state can produce different student responses due to mood, distraction, or context, so outcomes aren't guaranteed
Episodic vs Sequential	Sequential	Every decision (what to teach next, what difficulty to pick) depends on the accumulated history of past interactions; it is not a series of independent, memoryless episodes
Static vs Dynamic	Dynamic	The student's own knowledge changes as they learn, curriculum content can be updated, and other actors (a teacher assigning new work) can change the environment while the agent is reasoning
Discrete vs Continuous	Mostly discrete	Topics, quiz questions, difficulty tiers, and language choices are all finite/discrete sets; the one continuous

		element is the raw 0-1 mastery score, which the agent typically discretizes into bands for decision-making
Known vs Unknown	Known	The system operates within the well-defined curriculum, predefined subjects, grades, chapters, user roles, and educational rules. Although user questions vary, the domain knowledge and syllabus are known and structured.
Competitive vs Cooperative	Cooperative	The environment consists of students, teachers, parents, and administrators working together toward the common objective of improving student learning outcomes. There are no competing agents with conflicting goals.

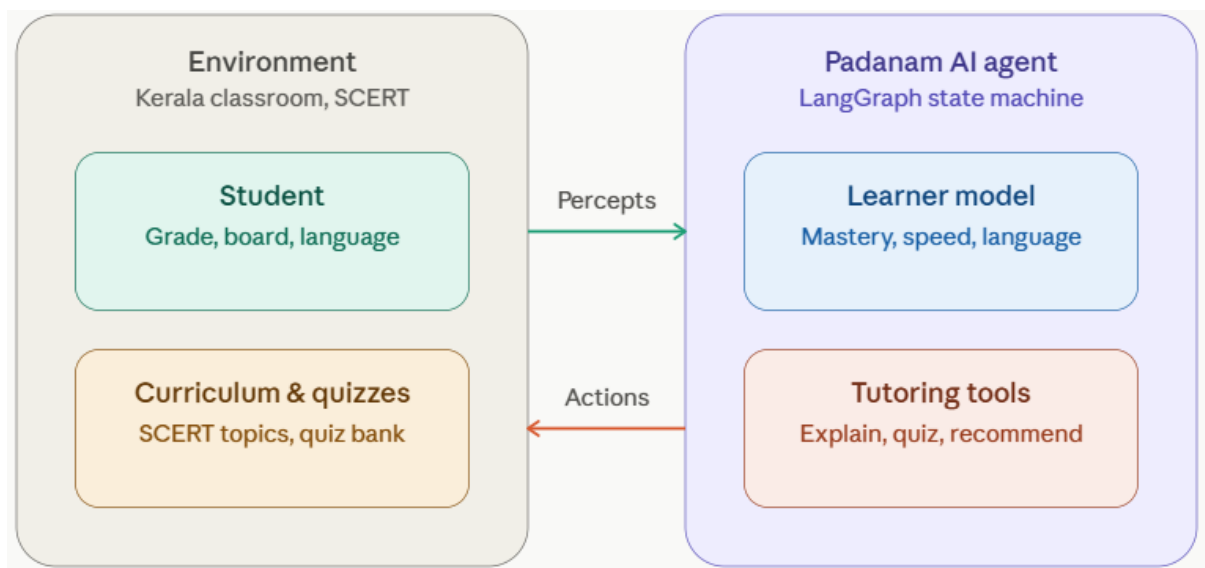
4. Rationality Measure

The agent is rational to the extent that, given its percept sequence (quiz results, response times, chat input, language choice) and built-in curriculum knowledge, it selects actions that maximize expected long-term mastery gain per unit of study time, while respecting the student's stated language preference and avoiding actions that would increase frustration. Rationality here is judged against outcomes accumulated over many sessions, not a single correct/incorrect action.

5. Innovation Point

Unlike generic chatbots or static repositories, Padanam AI diagnoses *why* an answer was wrong rather than just marking it incorrect, and switches its pedagogical strategy (analogy, simpler language, translation) instead of repeating a failed explanation. Its decision loop is modelled as an explicit, inspectable state graph rather than a single prompt-response call, making its reasoning debuggable and its adaptation genuinely tied to a persistent per-student model.

6. Diagram



7.Conclusion

Padanam AI demonstrates that a well-designed agent, with explicit state, sensors, and actuators, can turn a partially observable, stochastic, sequential learning environment into a tractable personalization problem. By grounding every recommendation in curriculum content and a persistent learner model, it offers a genuinely adaptive alternative to today's one-size-fits-all digital learning tools in Kerala.

8.Simulation

<https://padanam-ai.vercel.app/>

9.Screenshots of full code and output

The image displays two screenshots of the Padanam AI website. The top screenshot shows the main landing page with the title "Personalized Tutoring Grounded in Kerala SCERT Syllabus". The page features a dark blue background with white and teal text. At the top, there is a navigation bar with the Padanam AI logo, a language selector (EN | മലയാളം), a "Log In" link, and a "Student Signup" button. Below the navigation bar, there is a tagline "AI-Powered Adaptive Learning for SCERT Kerala Board". The main heading is "Personalized Tutoring Grounded in Kerala SCERT Syllabus". Below the heading, there is a paragraph describing the platform: "Padanam AI models every student's learning speed, language preference (English/Malayalam), and topic mastery — diagnosing exact misconceptions instead of just giving correct answers." At the bottom of the landing page, there are two buttons: "Start Learning Free →" and "Explore Demo Dashboards".

The bottom screenshot shows the "Create Your Padanam Account" registration form. The form has a dark blue background with white text. At the top, there is the Padanam AI logo and the title "Create Your Padanam Account". Below the title, there is a subtitle "Join the adaptive SCERT learning platform". The form contains several input fields: "FULL NAME" (with a placeholder "e.g. Anoop Kumar"), "EMAIL ADDRESS" (with a placeholder "student@example.com"), "PASSWORD" (with a placeholder "Minimum 6 characters"), "ACCOUNT ROLE" (with a dropdown menu showing "Student"), "LANGUAGE PREFERENCE" (with a dropdown menu showing "English"), and "SCERT CLASS / GRADE" (with a dropdown menu showing "Class 10 (SSLC Kerala)"). At the bottom of the form, there is a "Complete Registration →" button. Below the button, there is a link "Already registered? Sign in here".

Padanam AI

Select User Role

Choose a role below to enter the application immediately

INTERACTIVE LEARNING

Student

Access SCERT Kerala learning path, AI Tutor, courses & quizzes.

CLASSROOM INSIGHTS

Teacher

Monitor class progress, student performance analytics & curriculum.

PARENT PORTAL

Parent

Track child learning activity, weakness reports & study progress.

SYSTEM CONTROL

Admin

Manage system users, board settings & SCERT curriculum data.

Demo Simulation Mode • No credentials or password verification required

Padanam AI

SCERT KERALA STATE BOARD

EN | മലയാളം

Ask AI Tutor

A Anoop Kumar

CURRENT ROLE

Student Account

Dashboard

Subjects

Quizzes

Progress

Study Plan

Profile

Settings

SCERT Kerala Syllabus v10.4

Bilingual RAG • LangGraph AI Tutor active.

SCERT Kerala Grade 10 • Adaptive Mode

Welcome back, Anoop Kumar!

Padanam AI has updated your learning model. You have achieved **85% overall mastery** across your SCERT Kerala Class 10 subjects.

Ask AI Tutor Anything

Browse SCERT Chapters

OVERALL MASTERY

85%

Exponential Moving Average

TOTAL QUIZZES

3

Completed Diagnostic Items

WEAK TOPICS

1

Auto-flagged for remediation

AI LEARNING STYLE

Auto-Profiled

Balanced & Interactive

Steady pace with concept check quizzes

RECOMMENDED NEXT LESSON

AI Decision Engine

Reflection of Light

Physics Chapter 1 • SCERT Kerala State Board

TOPICS NEEDING REMEDIATION

Spaced Repetition Flag

Great job! No critical weak topics flagged right now.

പാദനം AI

SCERT KERALA STATE BOARD

മലയാളം | EN

എ.ഐ. ട്യൂട്ടറോട് ചോദിക്കൂ

A Anoop Kumar

CURRENT ROLE

Student Account

ഡാഷ്ബോർഡ്

വിഷയങ്ങൾ

ക്വിസ്സുകൾ

പുരോഗതി

പഠന പദ്ധതി

പ്രൊഫൈൽ

ക്രമീകരണങ്ങൾ

SCERT Kerala Syllabus v10.4

Bilingual RAG • LangGraph AI Tutor active.

SCERT Kerala Grade 10 • Adaptive Mode

വീണ്ടും സ്വാഗതം, Anoop Kumar!

Padanam AI has updated your learning model. You have achieved **85% overall mastery** across your SCERT Kerala Class 10 subjects.

Ask AI Tutor Anything

Browse SCERT Chapters

OVERALL MASTERY

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Padanam AI Tutor

SCERT Kerala Syllabus Grounded

LangGraph
Active



English

മലയാളം



● AI Tutor Mode: **English Explanation**

SCERT State Board



Hello! I am your Padanam AI tutor. Ask me anything about your SCERT Kerala syllabus, formulas, or concepts!

Pedagogy: standard

📝 Suggested Follow-ups:

Explain Wave Motion with a real-life analogy

സമാന്തര ഭ്രമണിയുടെ n-ാം പദം എങ്ങനെ കണ്ടെത്താം?

Give me a 2-minute diagnostic quiz

Ask about concepts, formulas, or SCERT syllabus...



Padanam AI

SCERT KERALA STATE BOARD

EN | മലയാളം

Ask AI Tutor

A Anoop Kumar

CURRENT ROLE

Student Account

Dashboard

Subjects

Quizzes

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SCERT Kerala Syllabus v10.4

Bilingual RAG + LangGraph AI

Tutor active.

SCERT Kerala Course Browser

Class 10 State Board Curriculum Chapters & Learning Outcomes

Search chapters or topics...

PhysicsMathematicsChemistry

CHAPTER 1

1 Topics

Wave Motion & Sound

Fundamentals of mechanical vs electromagnetic waves and sound propagation.

Topics in this chapter:

Transverse vs Longitudinal Waves

Study >

CHAPTER 2

1 Topics

Reflection & Refraction of Light

Laws of reflection, curved mirrors, refractive index, and Snell's Law.

Topics in this chapter:

Laws of Reflection & Mirrors

Study >

Padanam AI

SCERT KERALA STATE BOARD

EN | മലയാളം

Ask AI Tutor

A Anoop Kumar

CURRENT ROLE

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SCERT Kerala Syllabus v10.4

Bilingual RAG + LangGraph AI

Tutor active.

Back to SCERT Course Browser

SCERT KERALA CLASS 10 LESSON

Bilingual English / Malayalam

Transverse vs Longitudinal Waves

Curriculum BreakdownDiagnostic QuizAsk AI Tutor

Core Textbook Concept

Wave motion is a periodic disturbance propagating through a medium. In Transverse Waves, particles vibrate perpendicular to wave propagation (e.g. water ripples, light). In Longitudinal Waves, particles vibrate parallel to wave propagation (e.g. sound waves in air).

SCERT Learning Outcomes

LO-PHY10-1.1: Differentiate transverse and longitudinal waves based on particle vibration direction.

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Ask AI Tutor

A Anoop Kumar

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SCERT Kerala Syllabus v10.4

Bilingual RAG + LangGraph AI

Tutor active.

Back to SCERT Course Browser

SCERT KERALA CLASS 10 LESSON

Bilingual English / Malayalam

Laws of Reflection & Mirrors

Curriculum BreakdownDiagnostic QuizAsk AI Tutor

SCERT CLASS 10 QUIZ

Laws of Reflection & Mirrors

2 Questions

1. According to the Law of Reflection, if the angle of incidence is 35°, what is the angle of reflection?

A. 55°B. 35°C. 70°D. 90°

2. When light travels from air into glass, why does it bend towards the normal line?

A. Glass has lower optical density than air.B. Glass has higher optical density and slows down the light.

EN

മലയാളം

Ask AI Tutor

AAnoop Kumar

CURRENT ROLE

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SCERT Kerala Syllabus v10.4

Bilingual RAG + LangGraph AI Tutor active.

SCERT Diagnostic Quiz Center

Test your understanding and receive instant misconception diagnosis

?

Ready for a Diagnostic Quiz?

Test your understanding of Wave Motion & Energy Transmission with AI-generated SCERT questions.

Start Quiz Now

EN

മലയാളം

Ask AI Tutor

AAnoop Kumar

CURRENT ROLE

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SCERT Kerala Syllabus v10.4

Bilingual RAG + LangGraph AI Tutor active.

SCERT Diagnostic Quiz Center

Test your understanding and receive instant misconception diagnosis

SCERT CLASS 10 QUIZ

Transverse vs Longitudinal Waves

3 Questions

1. In a transverse wave, in which direction do the particles of the medium vibrate relative to the direction of wave motion?

A. Parallel to wave motion

B. Perpendicular to wave motion

C. In circular motion

D. Particles do not vibrate

2. Which of the following is an example of an electromagnetic wave?

A. Sound wave

B. Water surface wave

C. Light wave

D. Ultrasonic wave

3. What is the relationship between wave velocity (v), frequency (f), and wavelength (λ)?

EN

മലയാളം

Ask AI Tutor

AAnoop Kumar

Dashboard

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SCERT Kerala Syllabus v10.4

Bilingual RAG + LangGraph AI Tutor active.

Topic Mastery Heatmap (0 - 100%)

100%

75%

50%

25%

0%

Transverse vs Long...

Laws of Reflection...

Arithmetic Sequenc...

Mastered Concepts

Transverse vs Longitudinal Waves

76%

Laws of Reflection & Mirrors

88%

Arithmetic Sequences & nth Term Formula

92%

Remediation Needed

Padanam AI

SCERT KERALA STATE BOARD

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Ask AI Tutor

AAnoop Kumar

Dashboard

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SCERT Kerala Syllabus v10.4

Bilingual RAG + LangGraph AI

Tutor active.

ACTIVE ROADMAP

Regenerate with AI

D1

Review & Remediation: Wave Motion & Sound

Target Session: 45 mins

D2

Practice Quiz & Misconception Review: Wave Formulae ($v = f\lambda$)

Target Session: 30 mins

D3

New Topic: Laws of Reflection & Ray Diagrams

Target Session: 60 mins

D4

Malayalam/English Concept Breakdown & Formulas

Target Session: 45 mins

D5

Mathematics: Arithmetic Sequences (Common Difference d)

Target Session: 60 mins

D6

Full Topic Mastery Quiz & Performance Heatmap Check

Target Session: 45 mins

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SCERT KERALA STATE BOARD

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Ask AI Tutor

AAnoop Kumar

CURRENT ROLE

Student Account

Dashboard

Subjects

Quizzes

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Profile

Settings

SCERT Kerala Syllabus v10.4

Bilingual RAG + LangGraph AI

Tutor active.

User Profile

Account metadata, grade level, and state board settings

A

Anoop Kumar

Student Account • Padanam AI

EMAIL

STATE BOARD

SCERT Kerala State Board

CLASS / GRADE

Class 10 (SSLC)

LANGUAGE PREFERENCE

EN

Padanam AI

SCERT KERALA STATE BOARD

EN | മലയാളം

Ask AI Tutor

AAnoop Kumar

CURRENT ROLE

Student Account

Dashboard

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Profile

Settings

SCERT Kerala Syllabus v10.4

Bilingual RAG + LangGraph AI

Tutor active.

Platform Settings

Configure language, adaptive speed, and notifications

PREFERRED LEARNING LANGUAGE

English Medium

ADAPTIVE LEARNING SPEED MODEL

Standard Adaptive Speed

Save Preferences

Padanam AI

SCERT KERALA STATE BOARD

EN | മലയാളം

Ask AI Tutor

Dr. Sreedevi Nair

CURRENT ROLE

Teacher Account

Dashboard

Subjects

Teacher Portal

Profile

Settings

SCERT Kerala Syllabus v10.4

Bilingual RAG + LangGraph AI Tutor active.

Teacher Class Monitoring Dashboard

SCERT Class 10 Physics & Mathematics Class Progress Heatmap

ASSIGNED CLASS

Grade 10 - Section A

SCERT Model High School

CLASS AVERAGE MASTERY

75.0%

Exponential Moving Average

STUDENTS NEEDING SUPPORT

1 Student

Flagged by AI Tutor

Class Topic Mastery Roster

STUDENT NAME	GRADE	OVERALL MASTERY	WEAK TOPICS	RECENT QUIZ SCORE	STATUS
Anoop Kumar	Class 10	88%	None	90%	On Track
Meera Nair	Class 10	45%	2 Weak	50%	Intervention Recommended
Rahul Pillai	Class 10	72%	1 Weak	75%	On Track
Fatima Beevi	Class 10	95%	None	100%	On Track

Padanam AI

SCERT KERALA STATE BOARD

EN | മലയാളം

Ask AI Tutor

Rajesh Kumar

CURRENT ROLE

Parent Account

Dashboard

Parent Portal

Profile

Settings

SCERT Kerala Syllabus v10.4

Bilingual RAG + LangGraph AI Tutor active.

Parent Progress Portal

Jargon-free overview of your child's SCERT Kerala learning journey

Anoop Kumar

Class 10 • SCERT Kerala State Board

Overall Progress

82.5%

Recent Learning Activity:

Completed Quiz on Light Reflection & Refraction with 90% score.

Key Strengths & Mastery

- Excellent understanding of Arithmetic Sequences & Math formulas
- High retention of Physics Light laws and ray diagrams

Recommended Parent Support

- Needs a quick 15-minute review of Transverse Wave particle vibration direction

AI Encourage Note for Parent:

"Anoop is studying consistently! Encouraging him to take a 10-minute break between sessions will boost his focus further."

Padanam AI

SCERT KERALA STATE BOARD

EN | മലയാളം

Ask AI Tutor

System Administrator

CURRENT ROLE

Admin Account

Dashboard

Subjects

Teacher Portal

Parent Portal

Admin Management

Profile

Settings

SCERT Kerala Syllabus v10.4

Bilingual RAG + LangGraph AI Tutor active.

System Administration & Content Portal

Manage user roles, state boards, and vector database embeddings

TOTAL SYSTEM USERS

5

Student, Teacher, Parent, Admin

EDUCATION BOARDS

3

SCERT Kerala, CBSE, ICSE

ACTIVE SUBJECTS

3

Class 10 Physics, Maths, Chemistry

QUIZZES EVALUATED

4

With Misconception Analysis

Board Architecture Extensibility

Padanam AI models boards as dynamic foreign keys. Adding CBSE or ICSE textbooks requires zero schema modifications — simply upload new chapter chunks into the Chroma DB collection with the target board code tag.

SCERT_KERALA (Active)

Class 10 Wave Motion, Reflection, Arithmetic Sequences indexed.

CBSE (Extensible)

Ready for NCERT Class 10 textbook ingestion.

ICSE (Extensible)

Ready for CISCE Class 10 syllabus ingestion.

```

frontend > <> index.html > ...
1  <!doctype html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8" />
5      <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
6      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7      <title>Padanam AI (പാദനാം AI) - AI Learning Platform for SCERT Kerala</title>
8      <meta name="description" content="Adaptive, AI-powered personalized learning platform for SCERT Kerala" />
9      <link rel="preconnect" href="https://fonts.googleapis.com">
10     <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
11     <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800&family=Manrope:wght@300;400;500;600;700;800;900&family=Roboto:wght@300;400;500;600;700;800;900" rel="stylesheet">
12   </head>
13   <body class="bg-slate-900 text-slate-100 antialiased font-sans">
14     <div id="root"></div>
15     <script type="module" src="/src/main.jsx"></script>
16   </body>
17 </html>
18

```

```

() package.json App.jsx main.jsx X
frontend > src > main.jsx
1  import React from 'react';
2  import ReactDOM from 'react-dom/client';
3  import App from './App.jsx';
4  import './index.css';
5
6  ReactDOM.createRoot(document.getElementById('root')).render(
7    <React.StrictMode>
8      <App />
9    </React.StrictMode>,
10  );
11

```

```

frontend > src > App.jsx > ...
1  import React, { useState } from 'react';
2  import { BrowserRouter as Router, Routes, Route, Navigate } from 'react-router-dom';
3  import { AuthProvider, useAuth } from './context/AuthContext';
4  import { LanguageProvider } from './context/LanguageContext';
5
6  import Navbar from './components/common/Navbar';
7  import Sidebar from './components/common/Sidebar';
8  import AIChatDrawer from './components/chat/AIChatDrawer';
9
10 import LandingPage from './pages/LandingPage';
11 import LoginPage from './pages/LoginPage';
12 import RegisterPage from './pages/RegisterPage';
13 import StudentDashboard from './pages/StudentDashboard';
14 import CourseBrowser from './pages/CourseBrowser';
15 import LessonViewPage from './pages/LessonViewPage';
16 import QuizPage from './pages/QuizPage';
17 import AnalyticsPage from './pages/AnalyticsPage';
18 import StudyPlanPage from './pages/StudyPlanPage';
19 import ProfilePage from './pages/ProfilePage';
20 import SettingsPage from './pages/SettingsPage';
21 import TeacherDashboard from './pages/TeacherDashboard';
22 import ParentDashboard from './pages/ParentDashboard';
23 import AdminDashboard from './pages/AdminDashboard';
24 import NotificationsPage from './pages/NotificationsPage';
25
26 function ProtectedLayout({ children, onOpenChat }) {
27   const { user } = useAuth();
28   if (!user) return <Navigate to="/login" replace />;
29
30   return (
31     <div className="min-h-screen bg-slate-950 flex flex-col">
32       <Navbar onOpenChat={onOpenChat} />
33       <div className="flex flex-1 max-w-7xl w-full mx-auto">
34         <Sidebar />
35         <main className="flex-1 p-6 overflow-y-auto">
36           {children}
37         </main>
38       </div>
39     </div>
40   );
41 }
42
43 export default function App() {
44   const [onOpenChat, setOnOpenChat] = useState(false);
45
46   return (
47     <AuthProvider>
48       <LanguageProvider>
49         <Router>
50           <Routes>
51             <Route path="/" element={<LandingPage />} />
52             <Route path="/login" element={<LoginPage />} />
53             <Route path="/register" element={<RegisterPage />} />
54             <Route path="/student" element={<StudentDashboard />} />
55             <Route path="/course/:id" element={<CourseBrowser />} />
56             <Route path="/lesson/:id" element={<LessonViewPage />} />
57             <Route path="/quiz/:id" element={<QuizPage />} />
58             <Route path="/analytics" element={<AnalyticsPage />} />
59             <Route path="/study-plan" element={<StudyPlanPage />} />
60             <Route path="/profile" element={<ProfilePage />} />
61             <Route path="/settings" element={<SettingsPage />} />
62             <Route path="/teacher" element={<TeacherDashboard />} />
63             <Route path="/parent" element={<ParentDashboard />} />
64             <Route path="/admin" element={<AdminDashboard />} />
65             <Route path="/notifications" element={<NotificationsPage />} />
66           </Routes>
67         </Router>
68       </LanguageProvider>
69     </AuthProvider>
70   );
71 }
72

```

```

backend > app > main.py
1  import logging
2  from fastapi import FastAPI, Request, status
3  from fastapi.responses import JSONResponse
4  from fastapi.middleware.cors import CORSMiddleware
5  from app.core.config import settings
6  from app.core.logging import setup_logging, logger
7  from app.api.v1.router import api_router
8  from app.db.base import Base
9  from app.db.session import engine
10
11  setup_logging()
12
13  app = FastAPI(
14      title=settings.PROJECT_NAME,
15      openapi_url=f"{settings.API_V1_STR}/openapi.json",
16      description="Padanam AI - AI-Powered Personalized Learning Platform for SCERT Kerala Board Students",
17      version="1.0.0"
18  )
19
20  # CORS middleware for React frontend communication
21  app.add_middleware(
22      CORSMiddleware,
23      allow_origins=["*"],
24      allow_credentials=True,
25      allow_methods=["*"],
26      allow_headers=["*"],
27  )
28
29
30  @app.on_event("startup")
31  async def startup_event():
32      logger.info("Initializing database tables...")
33      async with engine.begin() as conn:
34          await conn.run_sync(Base.metadata.create_all)
35      logger.info("Database initialization complete.")
36
37

```

```

backend > app > main.py
31  async def startup_event():
36
37
38  # Centralized Exception Handler
39  @app.exception_handler(Exception)
40  async def global_exception_handler(request: Request, exc: Exception):
41      logger.error(f"Global unhandled exception on {request.url.path}: {exc}", exc_info=True)
42      return JSONResponse(
43          status_code=status.HTTP_500_INTERNAL_SERVER_ERROR,
44          content={
45              "detail": "An internal server error occurred.",
46              "error_code": "INTERNAL_SERVER_ERROR",
47              "path": request.url.path
48          }
49      )
50
51
52  app.include_router(api_router, prefix=settings.API_V1_STR)
53
54
55  @app.get("/")
56  async def root():
57      return {
58          "project": settings.PROJECT_NAME,
59          "version": "1.0.0",
60          "docs_url": f"{settings.API_V1_STR}/docs",
61          "status": "Running"
62      }
63

```

backend > app > services > llm_provider.py > ...

```
1  import json
2  import logging
3  from typing import Optional, Dict, Any, List
4  from app.core.config import settings
5
6  logger = logging.getLogger("padanam_ai.llm")
7
8
9  class LLMPProvider:
10     def __init__(self):
11         self.provider_type = settings.LLM_PROVIDER.lower()
12         logger.info(f"Initialized LLMPProvider with mode: '{self.provider_type}'")
13
14     async def generate_response(
15         self,
16         prompt: str,
17         system_prompt: Optional[str] = None,
18         lang: str = "en"
19     ) -> str:
20         if self.provider_type == "openai" and settings.OPENAI_API_KEY:
21             try:
22                 import httpx
23                 async with httpx.AsyncClient(timeout=30.0) as client:
24                     headers = {
25                         "Authorization": f"Bearer {settings.OPENAI_API_KEY}",
26                         "Content-Type": "application/json"
27                     }
28                     messages = []
29                     if system_prompt:
30                         messages.append({"role": "system", "content": system_prompt})
31                     messages.append({"role": "user", "content": prompt})
32
33                     payload = {
34                         "model": "gpt-4o-mini",
35                         "messages": messages,
36                         "temperature": 0.7
37                     }
```

backend > app > services > llm_provider.py > ...

```
9  class LLMPProvider:
14     async def generate_response(
37         )
38         resp = await client.post("https://api.openai.com/v1/chat/completions", headers=headers, json=payload)
39         if resp.status_code == 200:
40             data = resp.json()
41             return data["choices"][0]["message"]["content"].strip()
42         except Exception as e:
43             logger.warning(f"OpenAI call failed ({e}), falling back to local engine.")
44
45         elif self.provider_type == "gemini" and settings.GEMINI_API_KEY:
46             try:
47                 import httpx
48                 async with httpx.AsyncClient(timeout=30.0) as client:
49                     url = f"https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash:generateContent?key={settings.GEMINI_API_KEY}"
50                     contents = []
51                     if system_prompt:
52                         contents.append({"role": "user", "parts": [{"text": f"System Instruction: {system_prompt}"}]})
53                     contents.append({"role": "user", "parts": [{"text": prompt}]}))
54
55                     payload = {"contents": contents}
56                     resp = await client.post(url, json=payload)
57                     if resp.status_code == 200:
58                         data = resp.json()
59                         return data["candidates"][0]["content"]["parts"][0]["text"].strip()
60             except Exception as e:
61                 logger.warning(f"Gemini call failed ({e}), falling back to local engine.")
62
63         # Local Rule-Based Fallback Engine (Guarantees zero-downtime demo/viva without API keys)
64         return self._local_heuristic_response(prompt, system_prompt, lang)
65
66     def _local_heuristic_response(
67         self,
68         prompt: str,
69         system_prompt: Optional[str] = None,
70         lang: str = "en"
71     ) -> str:
```

```

backend > app > services > llm_provider.py > ...
9 class LLMProvider:
66 def _local_heuristic_response(
69     system_prompt: Optional[str] = None,
70     lang: str = "en"
71 ) -> str:
72     lower_prompt = prompt.lower()
73     is_ml = (lang == "ml") or ("malayalam" in lower_prompt) or ("മലയാളം" in lower_prompt) or ("ml" in (system_prompt or "")
74
75     if is_ml:
76         if "wave" in lower_prompt or "sound" in lower_prompt or "തരംഗ" in lower_prompt:
77             return (
78                 "**തരംഗ ചലനം (Wave Motion - SCERT 10 Physics)**:\n\n"
79                 "ഊർജ്ജം ഒരു സമലതത്തിന് മറ്റൊരു സമലതേക്ക് മാധ്യമത്തിന്റേ കണികകളുടെ യഥാർത്ഥ വ്യതിയാനം ചെയ്തുകൊണ്ട് പ്രസരണം ചെയ്യുന്നു. \n\n"
80                 "1. **അനുപ്രസ്ഥ തരംഗങ്ങൾ (Transverse Waves)**: മാധ്യമത്തിന്റേ കണികകൾ തരംഗ ചലന ദിശയ്ക്ക് ലംബമായി വ്യതിയാനം ചെയ്യുന്നു. \n\n"
81                 "2. **അനുദൈർഘ്യ തരംഗങ്ങൾ (Longitudinal Waves)**: മാധ്യമത്തിന്റേ കണികകൾ തരംഗ ചലന ദിശയ്ക്ക് സമാന്തരമായി വ്യതിയാനം ചെയ്യുന്നു. \n\n"
82                 "പ്രധാന സൂത്രവാക്യം: തരംഗ വേഗം  $v = f \times \lambda$  (ഇവിടെ  $f$  തരംഗ ആവൃത്തിയും,  $\lambda$  തരംഗദൈർഘ്യവും). \n\n"
83             )
84         elif "light" in lower_prompt or "reflection" in lower_prompt or "refraction" in lower_prompt or "പ്രകാശം" in lower_prompt:
85             return (
86                 "**പ്രകാശ പ്രതിഫലനം (Reflection & Refraction of Light)**:\n\n"
87                 "ഒരു പ്രകാശരശ്മി മിനുസമാർന്ന പ്രതലത്തിൽ തട്ടി തിരിച്ചുവരുന്ന പ്രക്രിയയാണ് പ്രകാശ പ്രതിഫലനം. \n\n"
88                 "• **പ്രതിഫലന നിയമങ്ങൾ**:\n\n"
89                 "1. പതനകോണും (Angle of Incidence,  $i$ ) പ്രതിഫലനകോണും (Angle of Reflection,  $r$ ) തുല്യമായിരിക്കും. \n\n"
90                 "2. പതനരശ്മി, പ്രതിഫലനരശ്മി, പതനബിന്ദുവിലെ ലംബം എന്നിവ ഒരേ തലത്തിലായിരിക്കും. \n\n"
91             )
92         elif "arithmetic" in lower_prompt or "sequence" in lower_prompt or "math" in lower_prompt or "ശ്രേണി" in lower_prompt:
93             return (
94                 "**സമാന്തര ശ്രേണി (Arithmetic Sequences - SCERT 10 Mathematics)**:\n\n"
95                 "ഒരു പദത്തോടൊപ്പം ഒരു നിശ്ചിത സംഖ്യ (പൊതുവ്യത്യാസം,  $d$ ) കൂട്ടി അടുത്ത പദം കണ്ടെത്തുന്ന സംഖ്യകളുടെ ക്രമമാണ് അрифമിറ്റിക് ശ്രേണി. \n\n"
96                 "•  $n$ -ാം പദം ( $n$ -th term):  $a_n = a + (n - 1)d$  \n\n"
97                 "• ആദ്യ  $n$  പദങ്ങളുടെ തുക:  $S_n = \frac{n}{2} [2a + (n - 1)d]$  \n\n"
98             )
99         else:
100             return (
101                 "**പാദനം എ.ഐ.പി.ടി സഹായി (Padanam AI Tutor)**:\n\n"
102                 "നിങ്ങൾ ചോദിച്ച വിഷയത്തെക്കുറിച്ചുള്ള ലളിതമായ വിശദീകരണം: \n\n"
103                 "എസ്.സി.ഇ.ആർ.ടി സിലബസ് അനുസരിച്ച് അടിസ്ഥാന ആശയങ്ങൾ ക്രമമായി മനസ്സിലാക്കി പഠിക്കുന്നു."
104             )
111     )
112
113     if "wave" in lower_prompt or "sound" in lower_prompt:
114         return (
115             "**Wave Motion & Sound (SCERT Class 10 Physics)**:\n\n"
116             "Wave motion transfers energy through a medium without transferring physical matter. \n\n"
117             "1. **Transverse Waves**: Particle vibration is perpendicular to wave motion (e.g., ripples on water). \n\n"
118             "2. **Longitudinal Waves**: Particle vibration is parallel to wave motion (e.g., Sound waves in air). \n\n"
119             "Key Formula: Velocity  $v = f \times \lambda$  (where  $f$  is frequency and  $\lambda$  is wavelength)."
120         )
121     elif "light" in lower_prompt or "reflection" in lower_prompt or "refraction" in lower_prompt:
122         return (
123             "**Reflection and Refraction of Light (SCERT Class 10 Physics)**:\n\n"
124             "Light travels in straight lines. When light encounters a boundary between two optical media: \n\n"
125             "• **Laws of Reflection**: Angle of incidence equals angle of reflection ( $i = r$ ). \n\n"
126             "• **Refraction**: Bending of light when entering a medium with a different optical density, governed by Snell's Law. \n\n"
127         )
128     elif "arithmetic" in lower_prompt or "sequence" in lower_prompt or "math" in lower_prompt:
129         return (
130             "**Arithmetic Sequences (SCERT Class 10 Mathematics)**:\n\n"
131             "An arithmetic sequence is a sequence of numbers where the difference between consecutive terms remains constant ( $d$ ). \n\n"
132             "•  $n$ -th Term Formula:  $a_n = a + (n - 1)d$  \n\n"
133             "• Sum of first  $n$  terms:  $S_n = \frac{n}{2} (a + a_n)$  \n\n"
134         )
135
136     return (
137         "Padanam AI Learning Assistant: I am tracking your SCERT Kerala State Board curriculum progress. "
138         "Mastering this concept requires breaking it down into fundamental principles, applying real-world analogies, "
139         "and taking targeted quick quizzes to test your understanding."
140     )
141
142 llm_provider = LLMProvider()
143
144

```